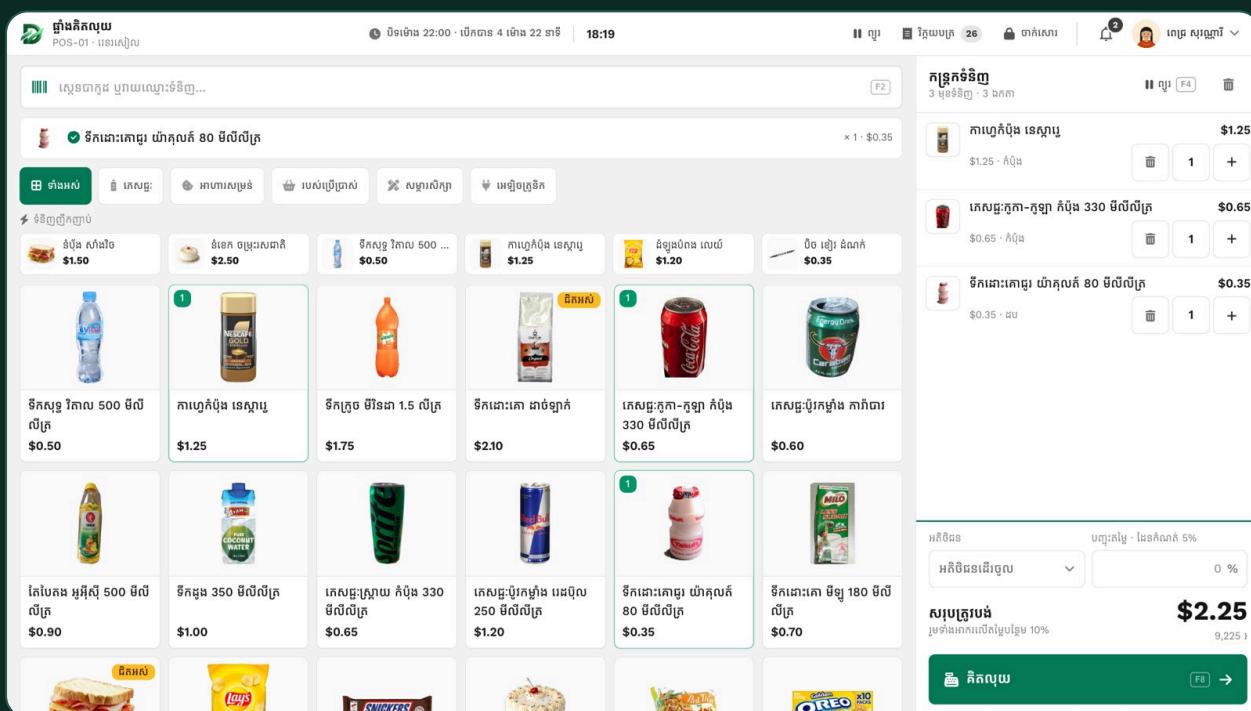




Detailed project report · October 2026

DIGITECHKH POS

A Khmer point-of-sale system for small retail shops in Cambodia: what it does, how it works, what the demo data shows, and what it needs before a real shop uses it.



Version
Prototype 1.0 · branch main

Date
6 October 2026

Prepared by
DIGITECHKH development team

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How to read this report. Sections 1–3 are for anyone deciding about the product. Sections 4–7 describe each screen and setting for the people who will use or test it. Sections 8–11 hold the business rules. Section 12 shows what the system produces from 14 days of realistic demo trading. Sections 13–16 are for the technical team.

Khmer labels are quoted «like this» exactly as they appear on screen. Money is in US dollars unless marked ៛ (riel). Screenshots come from the running prototype with its demo data.

1 Executive summary

DIGITECHKH POS is a working prototype of a point-of-sale system for a small Cambodian shop. Every screen is in Khmer. It covers a full trading day for three roles: the cashier sells, the shift manager controls, and the owner sees profit and sets the rules.

\$8,016 sales in 14 demo days	\$1,980 gross profit · 27.3% margin	2,702 receipts · \$2.97 average	79 drawer shifts on 3 registers
36 pages, all functional	52 products in 5 categories	10 staff: 6 cashiers, 3 managers, 1 owner	7 banks supported for KHQR

What the product does well

- **Fast counter work.** Scan, tap or search. Shortcuts F2 (search), F4 (hold) and F8 (pay) are printed on the buttons. A sale takes only a few taps, and a customer display mirrors the cart and the KHQR.
- **Cash you can trust.** The opening float is counted note by note and confirmed with a manager PIN. The till asks for safe drops above the drawer limit. Closing is a blind count with one recount, and the manager countersigns. Each register has one drawer with one cashier on it.
- **Exceptions are controlled and visible.** A discount above the cashier's limit, a void or a return needs a reason and a manager PIN. Managers can never approve their own sale. The Exceptions page compares each cashier with the team average and flags outliers.
- **Stock that can't be fiddled.** Quantities are never typed in; they are calculated from sales and recorded movements. Cashiers see only «ធុតិអស់» (low) and «អស់» (out), never numbers.
- **The owner sees the business.** Profit and margin by day, category, product and staff member; stock value at cost; shrinkage in dollars; payroll from real drawer hours; and a read-only audit log. Costs are never shown to cashiers or managers.
- **Made for Cambodia.** USD and KHR with a daily rate, change given as dollars plus riel rounded to 100, VAT-inclusive prices, KHQR to the shop's own bank accounts, an 80 mm receipt and an A4 Z-report.

What it is not yet

It is a front-end prototype. All data lives in one browser (localStorage), checks run in the browser, KHQR payments are confirmed by hand, and printing goes through the browser dialog. Section 16 sets out the work and the order to do it in.

Recommendation. (1) Fix the issues in section 15, which are small. (2) Run usability sessions with 3–5 real cashiers and a manager. (3) Build the backend and the Bakong integration, using this prototype as the specification. (4) Pilot in one shop for 2–4 weeks.

2 Scope, people and roles

The product serves one branch with three registers (POS-01 to POS-03). The demo shop, «DIGITECHKH» at «សាខាភណ្ណាល័យ ភ្នំពេញ», is open 24 hours in three shifts. Two cashiers work each shift. Both night cashiers are off on Tuesday, so a shift manager covers that night.

Role	Khmer	Responsibility	Lands on
Cashier	អ្នកគិតលុយ	Sells at one register; takes cash and KHQR; opens and closes their own drawer; requests voids and returns; makes safe drops	POS terminal
Shift manager	អ្នកគ្រប់គ្រងវេន	Approves exceptions; reviews closed shifts; plans the roster; runs the safe and stock; reads sales reports; covers a till on days off	Manager dashboard
Owner	ម្ចាស់ហាង	Sees profit, stock value and shrinkage; manages staff, pay, prices, costs, banks and shop rules; reads the audit log	Owner dashboard

Demo staff and default roster

Person	Role	Default shift	Day off	Discount limit	PIN
ចន្ទ មករា	Cashier	Morning 06:00–14:00	Sunday	5%	111111
សុខ ធារី	Cashier	Afternoon 14:00–22:00	Monday	5%	222222
លី សុភា	Cashier	Night 22:00–06:00	Tuesday	3%	333333
ពេជ្រ សុវណ្ណារី	Cashier	Afternoon 14:00–22:00	Thursday	5%	444444
គឹម វិសាល	Cashier	Morning 06:00–14:00	Wednesday	3%	555555
ឈឹម រតនា	Cashier	Night 22:00–06:00	Tuesday	3%	666666
សុខ រណ្ណា	Shift manager	Floor; covers a till	Saturday	–	246810
ម៉ៅ ស្រីនាង	Shift manager	Floor; covers a till	Sunday	–	135791
នួន សុខលី	Shift manager	Floor; covers a till	Wednesday	–	864201
ហេង ចាន់ថា	Owner	Not on the till	–	–	999999

Each cashier works 8 hours × 6 days = 48 hours a week, the legal maximum. Night hours (22:00–05:00) are paid at 200%.

Signing in

- Staff sign in with their email **or** phone number plus a password. The phone can be typed in any format (012 345 678, +855...).
- The 6-digit PIN is used only to approve something or to unlock the till.

- Five wrong tries lock the form for one minute. The error never says whether the account or the password was wrong. The last email or phone used is remembered.
- Each role lands on its own home page. Opening a page without the right role sends the person to the login page, and they come back to that page after signing in.
- The sidebar brand block switches views: the owner can switch to the manager view, and a manager to the cashier view.

DIGITECHKH
ប្រព័ន្ធគណនេយ្យ · សាខាភ្នំពេញ

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Figure 1. Login. One card with an illustration and the form; a live shift chip; demo accounts in the footer.

3 A day in the shop, step by step

This is the full cycle the prototype supports. Every step writes a record that the next role reads, so a manager sees a cashier's request appear without reloading the page.

#	Who	Step	What happens in the system
1	Cashier	Sign in and open shift	The page suggests a register (pinned → last used → first free). The cashier counts the float note by note in USD and KHR against the standard float (\$200 + ៛400,000). A manager confirms with a PIN. Only one open drawer is allowed per register and per person.
2	Cashier	Sell	Scan a barcode, tap a product or quick key, or search (F2). The cart shows the subtotal, discount, VAT and total in \$ and ៛.
3	Cashier	Take payment (F8)	Cash in USD, KHR or both; the change is split into \$ plus ៛ rounded to 100. Or KHQR: pick the bank, show the code (valid 5 minutes), confirm when the bank notification arrives. Mixed cash and KHQR is allowed. The receipt prints on 80 mm.
4	Cashier	Hold (F4)	Park a sale to serve the next customer, up to 5 per shift. Discarding a held sale needs a reason. Open held sales block the close.
5	Cashier → Manager	Discount above limit	The cashier picks a reason. A manager types their PIN on the cashier's screen. After 3 wrong PINs the dialog locks for 60 s and the attempt is logged.
6	Cashier → Manager	Void or return	Requested from the receipts list, with a reason. Approved on the spot with a PIN, or queued for the manager. Approval writes the stock movement; for a return, the manager chooses «ប្តូរស្តុកវិញ» (back to stock) or «ខូច» (damaged).
7	Cashier → Manager	Safe drop	A banner appears when the drawer passes \$500 or ៛2,000,000. The cashier records the drop; the manager counts it and confirms. An unconfirmed drop blocks the close.
8	Cashier	Lock the till	For a break; the cashier's own PIN unlocks it.
9	Cashier	Close shift	A 4-step blind count: information → count → variance → confirm. The expected amount stays hidden until the count is in; one logged recount is allowed. Then the A4 Z-report.
10	Manager	Review	Countersign the shift with a PIN; a note is required if the variance is over \$5. Or reopen it, with a reason. Large variances go to the dashboard and Exceptions.
11	Manager	Stock	Record a Stock In (supplier, invoice, quantity; no cost), adjustments with a reason, and the weekly blind stock count.
12	Owner	Review the business	Confirm the unit cost on each Stock In; check profit, cash variance by person, stock value and shrinkage; adjust prices, staff and rules; pay staff from the payroll page.

The four layers of the shift model

Layer	What it is	Who sets it
1 · Shift template	A time window: morning 06:00–14:00, afternoon 14:00–22:00, night 22:00–06:00 (overnight)	Owner
2 · Staff default	Each person's usual template and weekly day off	Owner / manager
3 · Roster per date	Generated from the defaults. On a day off a manager covers, marked «cover» with a reason. At most as many people as registers	Manager
4 · Drawer shift	One cashier, one register, one drawer: open → closed → reviewed (or reopened)	Cashier opens and closes; manager reviews

4 Cashier pages in detail

4.1 POS terminal «ផ្ទាំងលក់»

Area	Details
Product grid	Photos, 5 category tabs plus «ទាំងអស់», search (F2), a quick-keys row of 6 favourite products set by the manager. Paused products are hidden.
Stock badges	«ជិតអស់» (at or below the minimum) and «អស់» (out). No quantities. Selling at zero is allowed by default and logged as a stock mismatch.
Cart	Quantity steppers, remove line (logged), clear cart (logged), cart discount up to the cashier's own limit (3% or 5%) with a reason.
Totals	Subtotal → discount → VAT 10% (included in prices) → total, in \$ and ៛ at the rate of the day.
Payment	Cash keypad with note buttons for \$ and ៛; change split as \$ plus ៛ (or all ៛ when the customer paid in riel). KHQR with the bank chooser. Mixed tender.
Header	Shift chip, register, open-display button, safe-drop banner, lock, notifications.
Phones	The cart becomes a bottom sheet; targets stay 48 px.

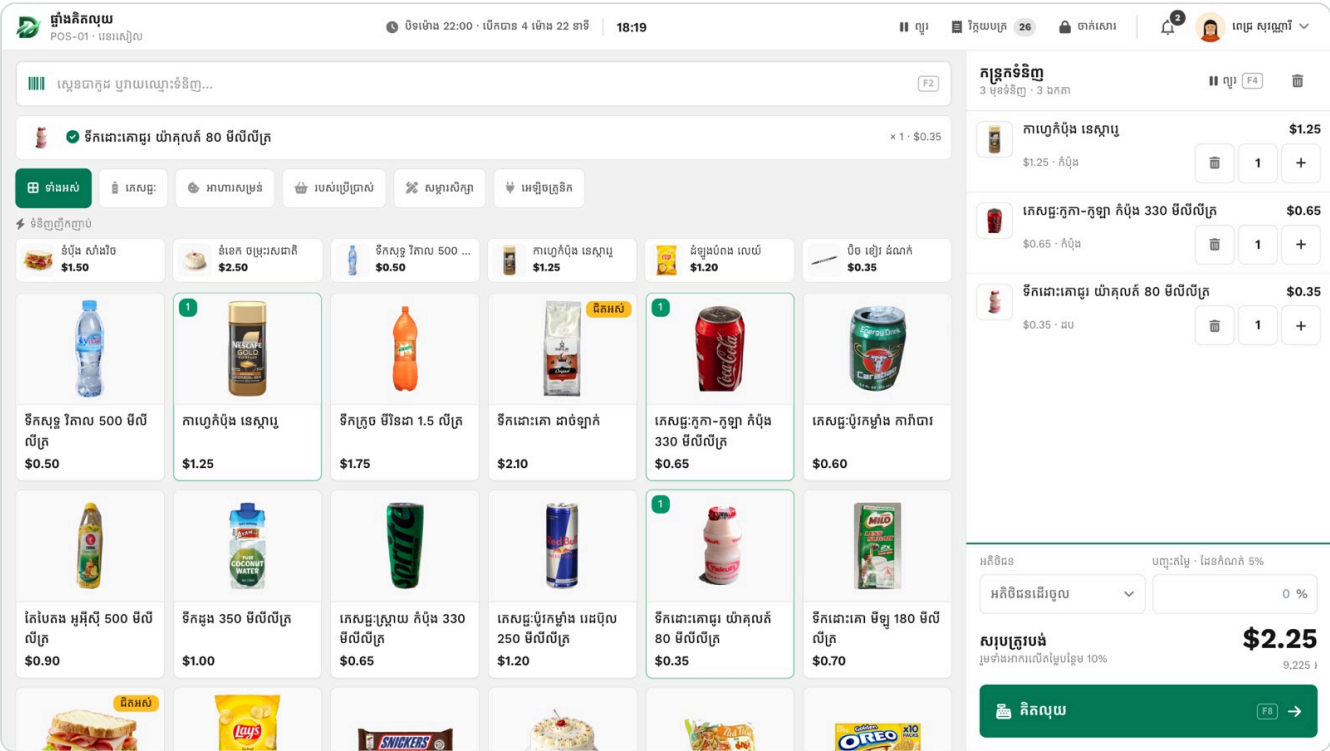


Figure 2. The terminal: grid, cart and pay button. Shortcuts are printed on the buttons.

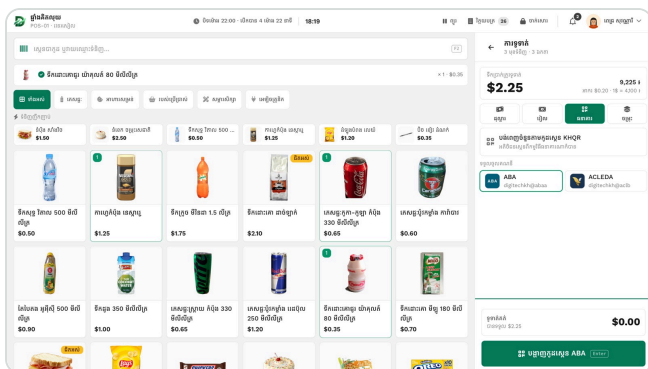


Figure 3. Payment: cash in two currencies, or KHQR to a chosen bank.

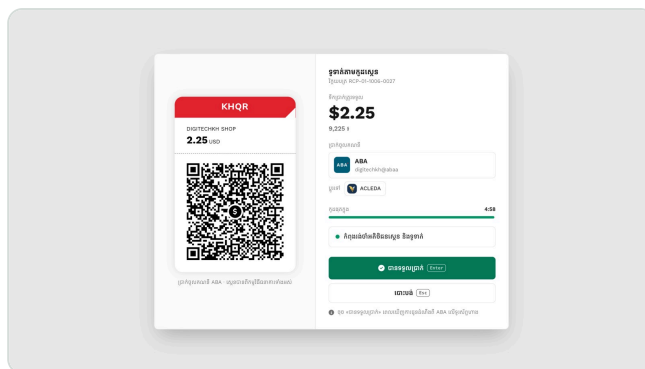


Figure 4. KHQR card in the Bakong layout: red header, name, amount, QR.

4.2 Customer display «អេក្រង់អតិថិជន»

Opened from the terminal header on a second screen. It is read-only and has five stages: **idle** (rotating shop messages, up to 5, set by the manager) → **sell** (lines and total) → **pay** → **khqr** (the same code the cashier sees) → **thanks** (change due). It shows nothing the customer shouldn't see: no cost, stock, PINs, reasons or staff names. The terminal republishes every 10 seconds; the display falls back to idle after 30 seconds without an update.

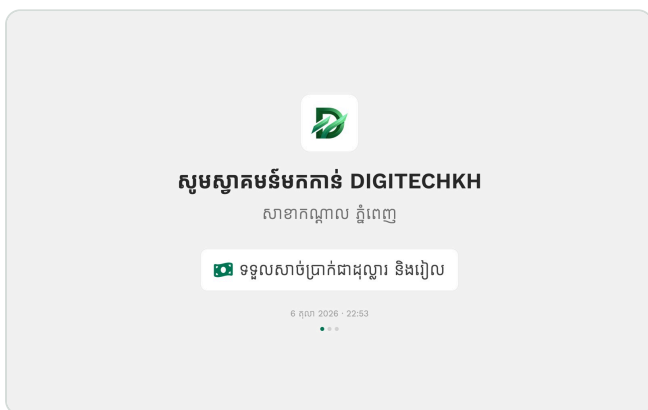


Figure 5. Customer display at idle.

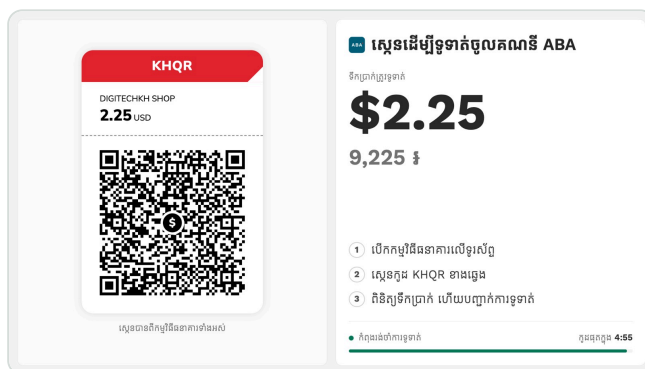


Figure 6. Customer display during KHQR payment.

4.3 Receipts «វិក្កយបត្រ»

Only the receipts of the cashier's **own current shift**; there is no history from other shifts on the cashier's device. 20 per page. Each row's : menu offers reprint (80 mm), request a void, and request a return (choose lines and quantity). Status chips show pending, approved or rejected; the row updates live when the manager decides.

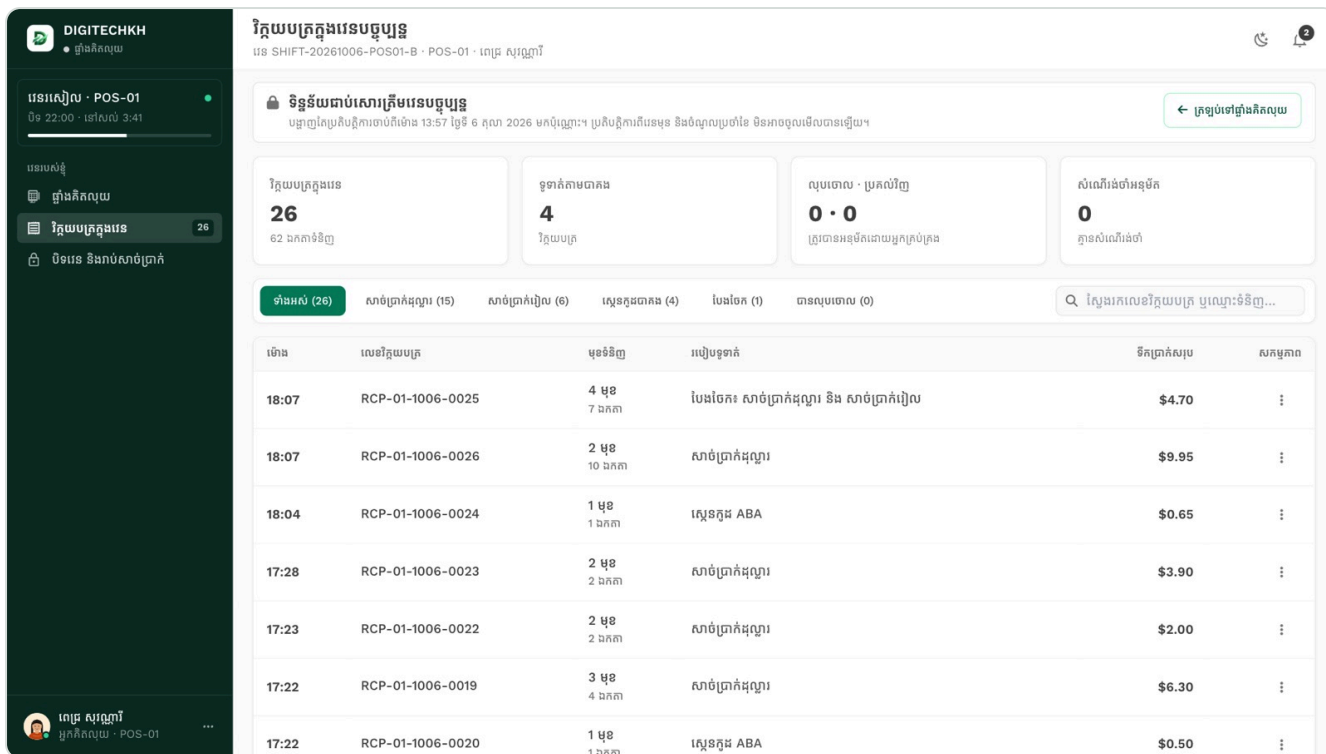


Figure 7. Shift receipts with void and return requests.

4.4 Open shift and close shift «បើកមេរ» / «បិទមេរ»

Page	Steps	Rules
Open shift	Register → float count (USD and KHR notes) → manager PIN	Standard float \$200 + ៛400,000; a difference needs a note. One open drawer per register and per person. With no rostered shift, a cover shift is offered.
Close shift	Information → blind count → variance → confirm	Expected cash hidden until counted; one recount, logged. Variance graded exact / small ($\leq$ \$5) / large. A large variance needs an explanation. Blocked by held sales or an unconfirmed drop. Prints the A4 Z-report.



5 Shift manager pages in detail

The sidebar shows badges for pending approvals, shifts awaiting review, unconfirmed drops and low stock. The policy line under the logo reminds the manager that they see every shift in the branch, never see costs, and can't approve their own requests.

Page	What is on it	Actions
Dashboard «ផ្ទាំងគ្រប់គ្រង»	KPIs (today's sales, receipts, average, pending); register cards (status, cashier, sales, cash in drawer against the limit, held sales); to-do list; sales by hour; payment donut; revenue by bank	Open a shift, an approval or a drop from the list
Approvals «សំណើរដាក់ចាំអនុម័ត»	Tabs pending / approved / rejected; type, receipt, cashier, amount, reason, waiting time	Approve (confirm + PIN) or reject (reason). The request page shows the receipt and the cashier's recent exception rates against the team
Shifts «វេន និងបញ្ជូនគិតលុយ»	All drawer shifts with status and variance per currency; awaiting review first	View: reconciliation per currency, count detail, explanation, exceptions, movements. Countersign, reopen, print the Z-report. An open shift shows as an X-report
Roster «កាលវិភាគវេន»	Week grid by template and date; covers, days off, weekly hours against 48	Assign, cover with a reason, pin to a register, move for a day
Cash «ចលនាសាច់ប្រាក់»	Safe balance in \$ and ៛; drops waiting; log of floats, drops, payouts, pay-ins, bank deposits	Confirm a drop (a mismatch needs a reason); create a movement (full page + PIN)
Stock «ស្តុកទំនិញ»	On-hand level per product, status, minimum, reorder list	Stock In (supplier, invoice, quantity, date; no cost); adjustment with a reason
Stock history «ប្រវត្តិស្តុក»	Every movement: sale, void, return, Stock In, adjustment, count, with who, when and why	Filter by type, product, person, date
Stock count «រាប់ស្តុក»	Blind counts; schedule weekly by default; overdue alert after 2 days	Count, one recount, differences go to Exceptions
Exceptions «ករណីមិនប្រក្រតី»	One row per cashier: voids, returns, overrides, line removals, cart clears, held sales discarded, wrong PINs, variance, off-hours sales; plus a by-approver table and stock exceptions	Click a cell for the event list
Sales report «របាយការណ៍លក់»	By day, hour, cashier, category, payment and bank, top products; gross, net, VAT, discount, receipts, average (selling prices only)	Print A4

Page	What is on it	Actions
Settings «ការកំណត់»	Rate of the day, quick keys, reason lists, customer display messages; the owner's rules shown read-only	Every save is audited

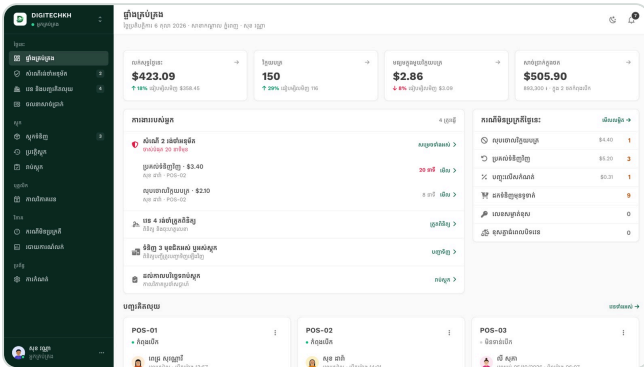


Figure 9. Manager dashboard.

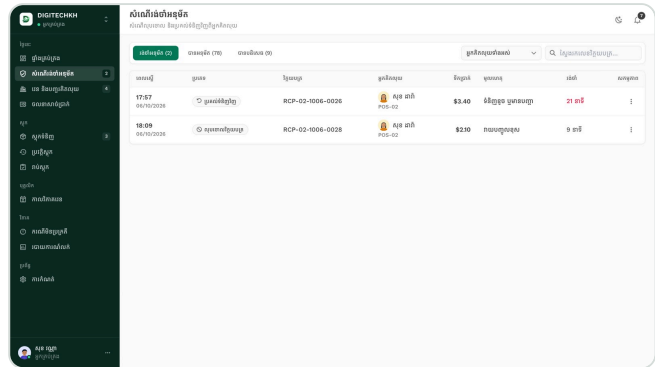


Figure 10. Approvals queue.

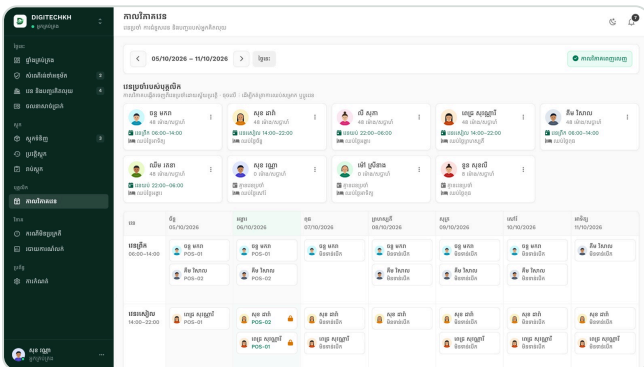


Figure 11. Weekly roster.

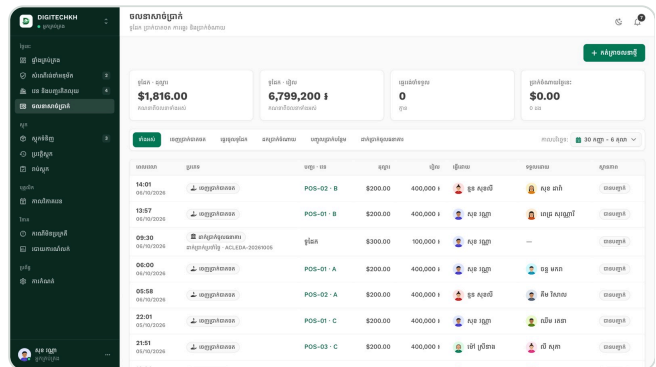


Figure 12. Cash office and safe.

6 Owner pages in detail

Page	What is on it	Actions
Dashboard «ទិដ្ឋភាពរួម»	Revenue, gross profit, margin, average basket and cash variance against the same weekday last week; 14-day profit chart; a needs-attention list (unconfirmed costs, large variances, low-margin products); results by staff; top products by profit	Every figure links to the list behind it
Revenue and profit «ចំណូល និងប្រាក់ចំណេញ»	Revenue ex-VAT, cost, gross profit and margin by day, category, product and staff	Date range; print A4
Stock In costs «បញ្ជាក់ថ្លៃដើមស្តុកចូល»	Each manager Stock In with lines and the last known cost	Confirm or edit unit costs; until then the report marks the cost «មិនទាន់បញ្ជាក់»
Stock value «តម្លៃស្តុកសរុប»	Value at cost by category and product, share of total	Search, print
Shrinkage «ការខាតបង់ស្តុក»	Loss at cost and retail by reason (damaged, expired, lost, count shortage), staff, category, product	Date range, drill-down
Staff «បុគ្គលិក និងតួនាទី»	Everyone with role, contact, status; profile page with shifts, timesheet and pay	:: edit profile, password and PIN, role, pay, deactivate. Safeguards: you can't deactivate yourself, the shop keeps one owner, and a person with an open shift stays active
Work time and payroll «ម៉ោងការងារ និងប្រាក់បៀវត្ស»	Per person: real hours split into normal / overtime / night, attendance, base pay, extras, deductions, net, status	Pay one or all, print the table, payslip
Products «ទំនិញ និងតម្លៃ»	Price, cost, unit profit, margin, 14-day units, minimum stock and reorder quantity	Change price or cost with a live margin preview; pause or resume
Settings «ច្បាប់ និងការកំណត់»	Every setting (section 7), including receiving banks and shift templates, with change history	Edit; every change audited
Audit log «កំណត់ហេតុសវនកម្ម»	499 entries in the demo: approvals, settings, staff, catalogue, costs, shift reviews, security events	Filter by type, person and date; read-only

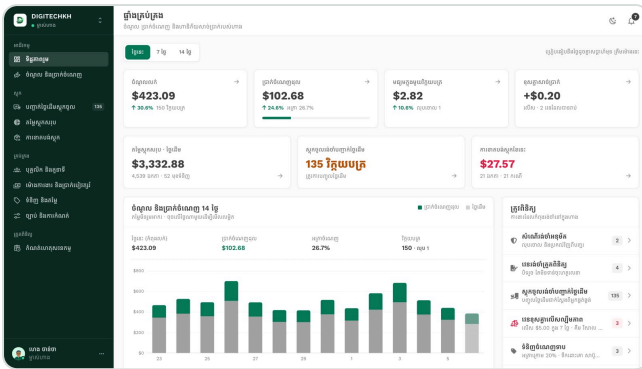


Figure 16. Owner dashboard.

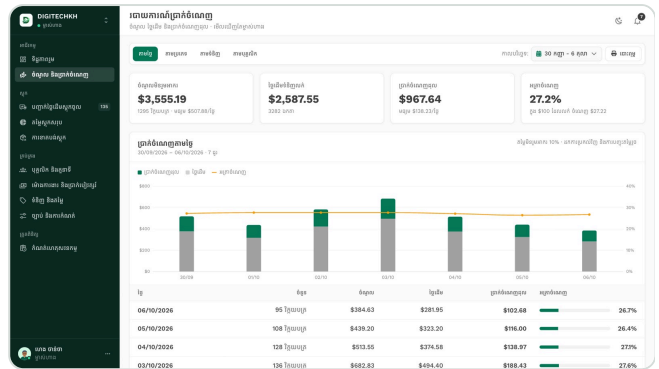


Figure 17. Revenue and profit report.

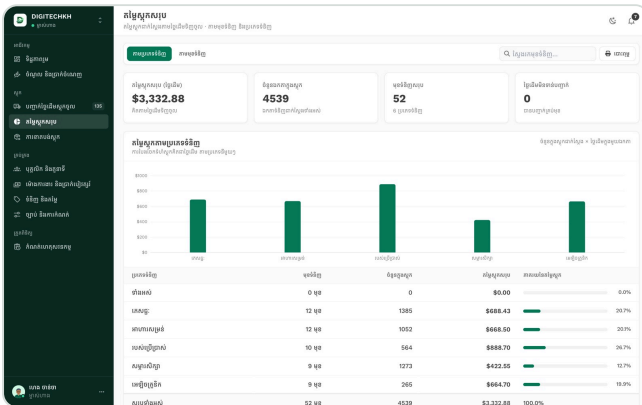


Figure 18. Stock value at cost (see issue 1 in section 15).

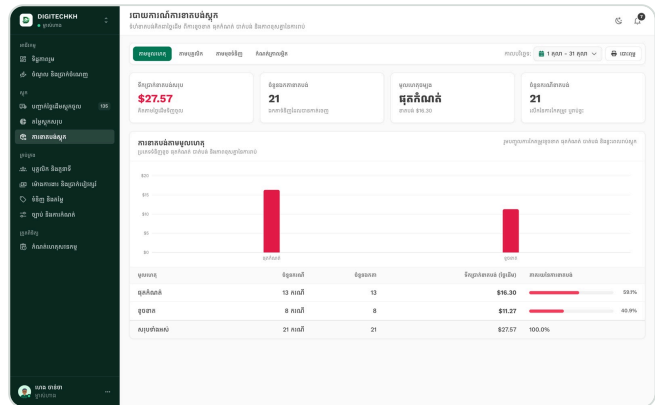


Figure 19. Shrinkage report.

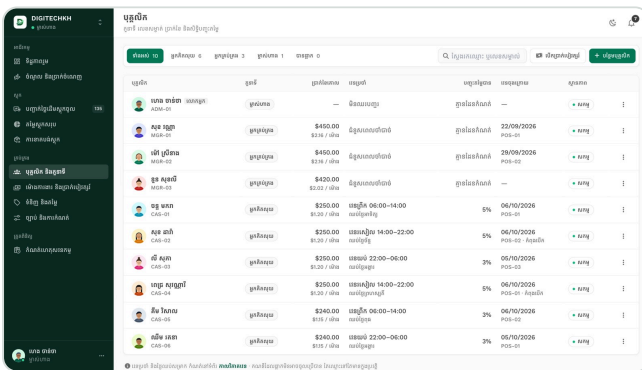


Figure 20. Staff and roles.

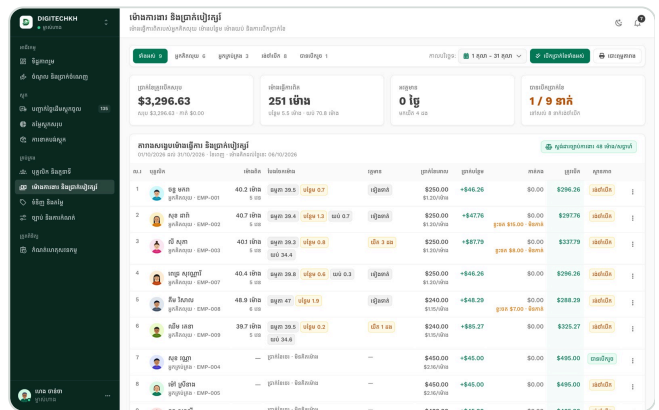


Figure 21. Work time and payroll for October.

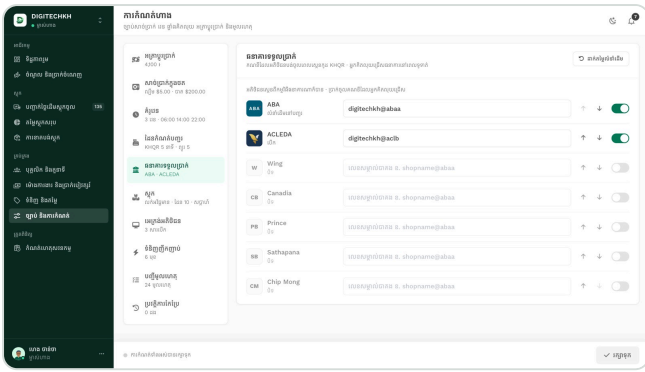


Figure 22. Receiving banks for KHQR.

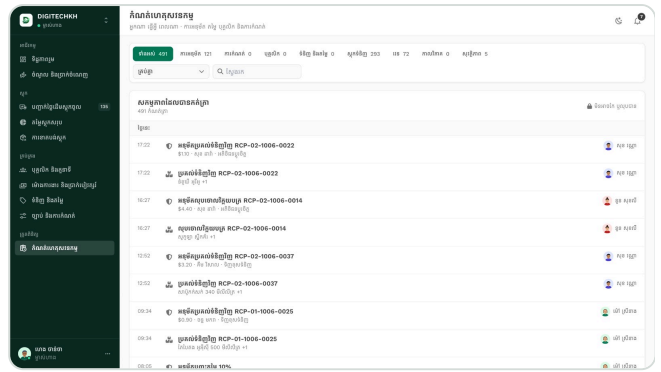


Figure 23. Audit log.

7 Settings reference

All settings live in one store with a change history (who, when, old → new). The manager edits the day-to-day ones. The rest the manager sees read-only and only the owner changes.

Setting	Khmer label	Default	Edited by
Rate of the day	អត្រាប្តូរប្រាក់ថ្ងៃនេះ	4,100 ៛ per \$ (stamped on every sale)	Manager
Official NBC rate (reference)	អត្រាផ្លូវការធនាគារជាតិ	not set	Manager
Quick keys	ទំនិញញឹកញាប់	6 products	Manager
Reason lists	បញ្ជីមូលហេតុ	void 4 · return 4 · discount 4 · payout 4 · hold discard 3 · cover 5	Manager
Customer display messages	សារនៅអេក្រង់អតិថិជន	3 messages (max 5), each with an icon and on/off	Manager
Variance tolerance	ល្បឹមភាពខុសគ្នាសាច់ប្រាក់	\$5.00	Owner
Drawer limit before a drop	ពិធានសាច់ប្រាក់ក្នុងថងក	\$500 / ៛2,000,000	Owner
Standard float	ប្រាក់បាតថងកស្តង់ដារ	\$200 / ៛400,000	Owner
KHQR code validity	សុពលភាពកូដស្កេនបាតង	300 seconds	Owner
Held sales per shift	ការលក់ប្តូរអតិបរមា	5	Owner
Discount limit per cashier	ដែនកំណត់បញ្ចុះតម្លៃ	3% or 5% per person	Owner
Shift templates	គំរូវេន	Morning 06–14, afternoon 14–22, night 22–06	Owner
Staff default shifts and days off	វេនលំនាំដើមរបស់បុគ្គលិក	See section 2	Owner
Receiving banks	ធនាគារទទួលប្រាក់	ABA and ACLEDA on; Wing, Canadia, Prince, Sathapana, Chip Mong off	Owner
Sell when stock shows zero	លក់បានពេលអស់ស្តុក	On (logged as a mismatch)	Owner
Adjustment limit	ដែនកំណត់កែតម្រូវស្តុក	10 units or \$50 at retail; above either goes to Exceptions	Owner
Stock count schedule	កាលវិភាគរាប់ស្តុក	Weekly	Owner

Default reason lists

List	Reasons
Void	រាយបញ្ចូលខុស · អតិថិជនប្តូរចិត្ត · ទូទាត់ខុសវិធី · ទំនិញខូច
Return	ទំនិញខូច ឬមានបញ្ហា · អតិថិជនប្តូរចិត្ត · ទិញខុសទំនិញ · ផុតកំណត់ប្រើប្រាស់
Discount	ទំនិញជិតផុតកំណត់ · អតិថិជនប្រចាំ · កញ្ចប់ខូចបន្តិច · ការផ្សព្វផ្សាយ
Payout	ទិញទឹកកក · ថ្លៃដឹកជញ្ជូន · សម្ភារសម្អាត · ចំណាយផ្សេងៗ
Hold discard	អតិថិជនមិនត្រឡប់មកវិញ · អតិថិជនលែងចង់ទិញ · បង្កើតខុស
Cover	ឈប់សម្រាក · ឈឺ · ប្តូរវេនគ្នា · ពេលមមាញឹក · ផ្សេងៗ
Stock adjustment	ខូចខាត · ផុតកំណត់ · បាត់ · រកឃើញវិញ · ប្រើក្នុងហាង (the first three count as shrinkage)

8 Controls, approvals and security

8.1 Permission matrix

Action	Cashier	Manager	Owner
Sell, hold, take cash and KHQR	Yes	Yes (when covering)	–
Remove a line or clear the cart before payment	Yes, logged	Yes	–
Discount up to own limit	Yes + reason	Yes	–
Discount above limit	No	PIN at the till	Yes
Void or return a completed sale	Request	Approve	Approve
Approve their own sale	–	Never	–
Issue the float, confirm drops, payouts	No	Yes	Yes
Close shift (blind count)	Yes	Yes (own drawer)	–
Countersign or reopen a shift	No	Yes	Yes
See stock quantities	Status only	Yes	Yes
Stock In, adjust, count	No	Yes (no cost)	Yes + cost
Rate of the day, quick keys, reasons	No	Yes	Yes
Cash rules, templates, limits, banks	No	Read-only	Yes
Cost, profit, margin, stock value, shrinkage \$	No	No	Yes
Staff, pay, prices, costs	No	No	Yes
Audit log	No	No	Read-only

8.2 Two ways to approve

On the spot. The terminal opens a dark dialog on the cashier's screen with the action, the receipt, a reason, the approver (never the cashier themselves) and a 6-digit PIN keypad. Three wrong PINs lock it for 60 s and log an «override denied» event. The audit entry is marked on-site.

Queued. A request raised from the receipts list while no manager is near waits in the approvals queue. The manager sees the receipt and the cashier's recent rates against the team, then approves with a PIN or rejects with a reason. The cashier's screen updates live.

8.3 What an approval changes

Decision	Effect
Void approved	The sale is marked voided and counts as zero in every report; all items go back to stock
Return approved	The returned lines are added to the sale; a cash refund lowers the expected cash; the stock goes back, or is written off as damaged (shrinkage against the selling cashier)
Discount override	The sale stores the approver; it counts in Exceptions as an override
Reopen shift	Status goes back to open, with who reopened it and why
Reject	A reason is required; the cashier is notified; the sale is unchanged

8.4 How the rules are enforced

- **Data separation by script.** Costs exist only in `admin-data.js`, loaded by owner pages only. Other shifts' history exists only in `manager-data.js`. The cashier's `data.js` never contains other shifts, so the cashier's lock is real, not just hidden.
- **Page guards.** `cashier/*` needs a session (an owner is sent to the owner home), `manager/*` a manager or owner, `admin/*` an owner.
- **Traceability.** Voids, rejections, adjustments, covers, deactivations and pauses all need a reason; every record stores the person and the time; nothing can be deleted.
- **Patterns over single events.** Exceptions flag a cashier whose rate is 2× (amber) or 3× (red) the team average, plus sales outside shift hours and wrong-PIN attempts. A by-approver table shows managers who approve everything.
- **Accounts.** Phone and email are unique per active person; you can't deactivate yourself; the shop always keeps one owner.

Limitation. In the prototype, all checks run in the browser, and seed passwords and PINs ship in the page source. They demonstrate the behaviour; they are not a security boundary until a server enforces them (section 16).

9 Money, currency and payment rules

Rule	How it works
Prices include VAT	Shelf prices include 10% VAT. A sale breaks down as subtotal → down payment → special discount → VAT 10% → grand total; net = total ÷ 1.10.
Two currencies	Every amount shows in \$ and ₭. The manager sets the daily rate; every sale stores the rate it used, so old receipts never change.
Change	Whole dollars plus the rest in riel rounded to the nearest ₭100. A customer who paid only in riel gets all-riel change.
Price snapshot	Each sale line stores its price; a later price change never alters old receipts or reports.
Expected cash	Per currency: float + cash sales – change – cash refunds – drops – payouts + pay-ins.
Variance	Counted – expected, converted to \$ at the shift rate: exact (0), small ($\leq$ tolerance, \$5) or large ($>$ \$5, needs an explanation and a manager note).
Profit (owner only)	Revenue ex-VAT after discounts and returns, minus unit cost × quantity. Voided sales count as zero.
KHQR	A valid EMV KHQR string (tag 29 account, 52 merchant category, 63 CRC16) for the chosen bank account. Valid 5 minutes, then regenerate. The cashier confirms on the bank notification. The sale stores the bank, so reports split revenue per bank.

Worked example

A customer buys a Nescafé can (\$1.25) and a Lay's bag (\$1.20) and pays with a \$5 note at ₭4,100. Total \$2.45 (VAT included: net \$2.23 + VAT \$0.22) · change \$2.55 → **\$2 + ₭2,300** ($0.55 \times 4,100 = 2,255$, rounded to 2,300; the shop gives ₭45 extra, which is recorded as rounding).

Printing

- **Receipt:** 80 mm thermal layout with the shop name, branch, VAT TIN, lines, VAT breakdown, tender, change in \$ and ₭, the cashier and the register. A VAT-registered credit customer gets a tax invoice.
- **Z-report:** A4, with reconciliation per currency, counts by note, exceptions and two signature blocks (the cashier at close, the manager at review).

10 Stock rules

- **Quantity is derived, never stored or edited:** on hand = opening balance + sales since the opening (read from receipts) + recorded movements.
- **Movements:** void (+), return (+, or a damaged write-off), Stock In (+), adjustment ($\pm$, reason required), count ($\pm$ difference). Each records who, when, the quantity and the reason.
- **Status:** low = on hand $\leq$ the product's minimum; out = 0 or below. The reorder list suggests the product's reorder quantity.

- **Exceptions:** selling at zero (stock mismatch), an adjustment above 10 units or \$50 at retail, a count difference, a count more than 2 days overdue.
- **Costs:** a manager Stock In has no cost; the owner confirms it. Until then the last known cost is used and marked unconfirmed.

11 Work time and payroll rules

Pay comes from **real** time, meaning when the drawer was actually opened and closed, compared with the roster. Every amount can be traced to hours, and every hour to a dated shift.

Rule	Value
Normal hours	8 per day, 48 per week (Monday–Sunday); 26 working days a month, so hourly rate = base ÷ 208
Overtime	150% by day, 200% at night
Night work 22:00–05:00	200% (to confirm with an accountant)
Work on the weekly day off	200%
Late or early	Counted beyond a 10-minute grace
Attendance bonus	Kept with no absence and at most 3 late arrivals
Cash shortages	Shown on the payslip but not deducted until the owner decides
Managers	Fixed monthly salary; hours not tracked for pay
Owner	Not on the payroll

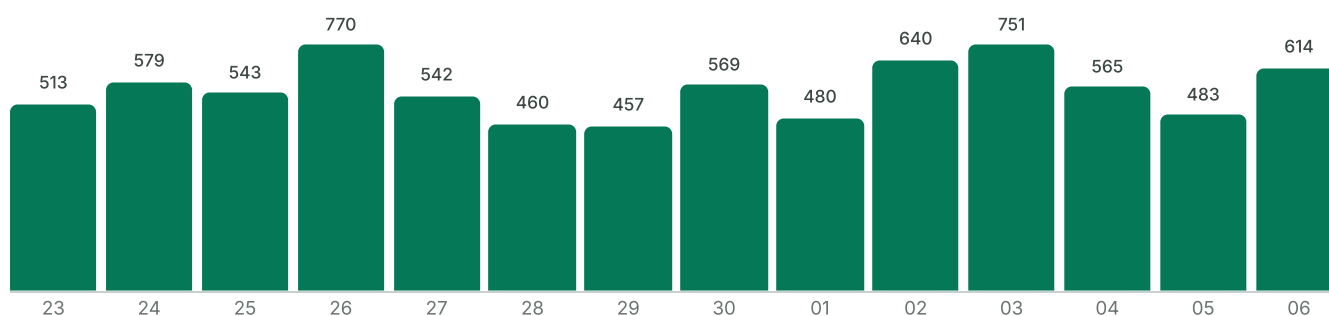
12 Demo data results (14 days)

The prototype generates 14 days of trading (2026-09-23 to 2026-10-06) from the roster, following a busy-hour curve, product popularity and a realistic tender mix. The figures below are what the owner and manager pages show. They prove that the reports add up; they are **not** forecasts for a real shop.

12.1 Headline figures

Measure	Value	Measure	Value
Gross sales (VAT incl.)	\$8,016.26	Receipts (not voided)	2,702
Returns	\$50.59	Voided receipts	40
Net sales	\$7,965.67	Items sold	6,934
VAT collected	\$728.75	Average receipt	\$2.97
Discounts given	\$22.44	Items per receipt	2.57
Revenue ex-VAT	\$7,241.52	Average sales per day	\$572.59
Cost of goods	\$5,261.44	Best day	Sat 26/09 · \$769.77
Gross profit	\$1,980.08	Weakest day	Tue 29/09 · \$457.13
Gross margin	27.3%	Product margins range	17.5% – 47.2%

12.2 Sales per day (\$)

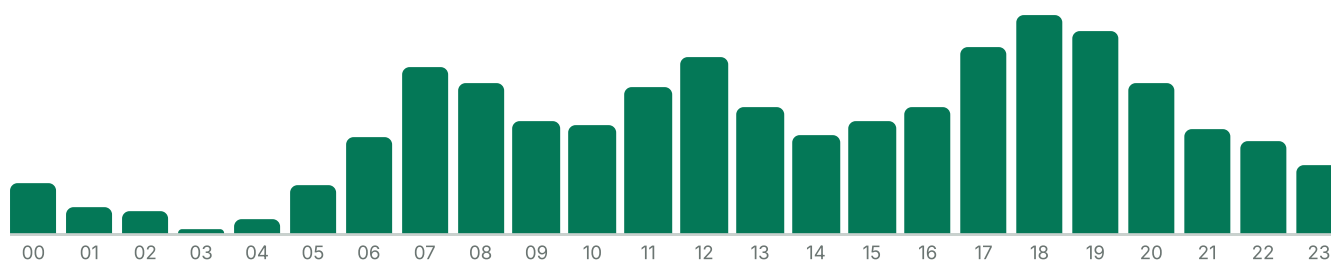


Day	Receipts	Sales	Gross profit	Margin	Cash variance
Wed 23/09	159	\$512.92	\$122.07	26.2%	\$-23.22
Thu 24/09	189	\$579.27	\$143.99	27.3%	\$-0.74
Fri 25/09	203	\$543.46	\$135.79	27.5%	\$0.80
Sat 26/09	258	\$769.77	\$192.89	27.6%	\$1.38
Sun 27/09	180	\$542.28	\$138.79	28.1%	\$-15.49
Mon 28/09	167	\$459.52	\$115.21	27.6%	\$-0.25

Day	Receipts	Sales	Gross profit	Margin	Cash variance
Tue 29/09	183	\$457.13	\$115.67	27.8%	\$-0.30
Wed 30/09	186	\$568.83	\$140.60	27.2%	\$-2.35
Thu 01/10	160	\$479.66	\$120.21	27.6%	\$-15.35
Fri 02/10	206	\$639.98	\$160.75	27.6%	\$-9.87
Sat 03/10	237	\$751.11	\$188.43	27.6%	\$-0.93
Sun 04/10	194	\$564.91	\$138.97	27.1%	\$-5.62
Mon 05/10	162	\$483.12	\$116.00	26.4%	\$0.00
Tue 06/10	218	\$613.71	\$150.70	27.0%	\$-4.35

Cash variance is the sum over the day's counted shifts; a shortage over \$5 is shown in red.

12.3 Receipts by hour of day (14 days)



Peaks at 07:00–08:00, 12:00 and 17:00–19:00; quiet from 02:00 to 04:00. This matches a neighbourhood convenience shop.

12.4 Payment mix

Tender	Amount	Share
Cash in US dollars	\$3,768.84	47.0%
Cash in riel (shown in \$)	\$2,030.55	25.3%
KHQR · ABA	\$1,490.10 · 482 payments	18.6%
KHQR · ACLEDA	\$726.77 · 239 payments	9.1%
Total	\$8,016.26	100%

12.5 Categories: sales and stock

Category	Units sold	Sales ex-VAT	Share	Products	Units in stock	Stock value at cost
Drinks «ភេសជ្ជៈ»	2,995	\$2,421.36	33.1%	12	1,385	\$688.43
Snacks «អាហារសម្រន់»	1,852	\$2,045.55	28.0%	12	1,052	\$668.50
Household «របស់ប្រើប្រាស់»	600	\$1,357.50	18.6%	10	564	\$888.70
Stationery «សម្ភារសិក្សា»	1,304	\$781.14	10.7%	9	1,273	\$422.55
Electronics «អេឡិចត្រូនិក»	183	\$702.36	9.6%	9	265	\$664.70

Stock on hand: **4,539 units worth \$3,332.88 at cost**. Three products need attention: «ទឹកដោះគោ ដាច់ឡាក់» and «ទំប៉ុង សាំងរិច» are low, and «កាសស្តាប់ចម្រៀង» is out.

12.6 Best-selling products

#	Product	Units	Price	Margin
1	ទឹកស្អាត វ៉ាល 500 មីលីលីត្រ	565	\$0.50	34.0%
2	កាហ្វេកំប៉ុង ទេស្តាម្លៃ	409	\$1.25	25.2%
3	គេសឆ្នុះក្អក-ក្អាត កំប៉ុង 330 មីលីលីត្រ	336	\$0.65	28.9%
4	ប៊ិច ខៀវ ដំណក់	306	\$0.35	43.4%
5	ដំឡូងបំពង លេយ៍	297	\$1.20	26.7%
6	មីកញ្ចប់ មាម៉ា	284	\$0.40	28.5%
7	ស្បៀងកៅស៊ូសរសេរ 100 ទំព័រ	268	\$0.75	34.0%
8	ទឹកដោះគោ មីឡូ 180 មីលីលីត្រ	266	\$0.70	27.7%

12.7 Shifts and cash control

Measure	Value
Drawer shifts	79 (74 reviewed, 4 closed and awaiting review, 1 open now)
Counts exactly right	35 of 78 (44.9%)
Small variance ($\leq$ \$5)	35 (44.9%)
Large variance ($>$ \$5)	8 (10.3%); each one needs an explanation and a manager note
Net cash variance, 14 days	\$-76.30 (shortage), 0.95% of sales
Safe balance now	\$2,210 + \$7,529,500

12.8 Approvals

Type	Approved	Rejected	Pending
Void	46	9	1
Return	35	0	1

12.9 Exceptions by cashier (14 days)

Cashier	Shifts	Receipts	Sales	Voids	Returns	Overrides	Wrong PINs	Variance
ចន្ទ មករា	12	575	\$1,719.79	10	10	4	0	\$-13.38
សុខ ធារី	12	631	\$1,815.82	7	7	7	1	\$-16.20
លី សុភា	12	170	\$460.16	9	2	4	0	\$-17.66
ពេជ្រ សុវណ្ណារី	12	636	\$1,891.73	8	6	6	2	\$-22.17
គឹម វិសាល	12	560	\$1,630.04	4	9	9	2	\$-9.27
ឈឹម រតនា	12	156	\$468.70	2	1	1	0	\$2.39

Team averages per 100 receipts: voids 1.47 · returns 1.28 · overrides 1.14 · lines removed 5.50 · carts cleared 1.25 · held sales discarded 1.39 · wrong PINs 0.18.

Reading it. The night cashier «លី សុភា» shows 9 voids on only 170 receipts, a rate of 5.3 per 100, which is 3.6× the team average. This is the kind of pattern the page is designed to surface. Counts alone would hide it, because night shifts have far fewer sales.

12.10 Shrinkage (14 days)

Reason	Events	Units	Loss at cost
ផុតកំណត់	27	27	\$35.82
ខូចខាត	15	15	\$25.47
ខ្វះពេលរាប់ស្តុក	1	1	\$0.30
Total	43	43	\$61.59 (\$94.45 at retail)

12.11 Payroll, October to date

The payroll page for 1–31 October (as of the 6th) shows **\$3,296.63** due for 9 people: 251 real hours, of which 5.5 are overtime and 70.8 are night hours. One person has been paid and 8 are pending. Cash shortages (\$15, \$8 and \$7) are shown against three cashiers but not deducted (figure 21).

13 Architecture and data dictionary

The whole product is a static site: HTML, Tailwind (Play CDN), vanilla JavaScript, ECharts 5 for charts, Font Awesome and Iconify for icons, and the Kantumruy Pro font. There is no build step and no server. `vercel.json` publishes `src/`; every push to `main` redeploys. All paths are relative, so the site also works under a subpath.

~14.6k

lines of HTML in 36
pages

~10.0k

lines of shared
JavaScript

~3.7k

lines of CSS

44

commits, 3–6 Oct 2026

13.1 Scripts and load order

Script	Lines	Loaded by	Holds
<code>ui-components.js</code>	2,291	All	Toasts, confirm, reason prompt and picker, option and form dialogs, manager-PIN override, PIN confirm, cash counter, number pad, pagination, list memory
<code>data.js</code>	2,551	All	Shop, catalogue, staff, PINs, settings, shift model, live records, money maths, stock derivation, KHQR payload, receipt and Z-report HTML, page guards, demo generators
<code>manager-data.js</code>	1,279	Manager, owner	14-day history, merged views, overlays, exceptions, aggregates, safe balance, notifications
<code>admin-data.js</code>	1,015	Owner	Unit costs, profit maths, Stock In costs, stock value, shrinkage, staff and catalogue edits, pay rules and payroll, audit trail
<code>portal.js</code>	1,548	All	Sidebar, header, view switcher, notifications, mobile drawer, : menus, date range, custom select, stepper
<code>settings-page.js</code>	904	Settings pages	Settings UI, sections filtered by role
<code>staff-actions.js</code>	344	Staff pages	Edit profile, password and PIN, role, pay, deactivate
<code>charts.js</code>	98	Chart pages	<code>posChart()</code> ECharts wrapper

Order: `ui-components` → `data` → `manager-data` → `admin-data` → `portal` → `charts` → page-specific → inline script. Cashier pages load neither `manager-data` nor `admin-data`; manager pages skip `admin-data`.

13.2 Browser storage keys

Key	Holds
<code>pos_session</code>	Signed-in person
<code>pos_settings</code>	Setting values and their change history

Key	Holds
pos_roster	Roster changes per date and template
pos_shifts · pos_shift_sales · pos_held_sales	Live drawer shifts, their sales, parked sales
pos_approvals · pos_cash_movements · pos_events	Requests, floats, drops, payouts, the event log
pos_terminal_lock	Locked till
pos_pins · pos_passwords · pos_login_guard · pos_last_login	Reset PINs and passwords, failed-login lock, last identifier
pos_stock_opening · pos_stock_moves · pos_stock_counts · pos_stock_costs	Stock opening time, movements, counts, confirmed Stock In costs
pos_overlay	Manager decisions on generated history
pos_staff · pos_catalog · pos_costs · pos_admin_log	Owner edits to staff, prices, costs, and their log
pos_cfd	Customer-display snapshot
sessionStorage pos_cart · pos_pending_khqr	Current cart, KHQR in progress

13.3 Main records

Record	Key fields
Sale	id, time, shiftId, register, cashierId, items (sku, qty, price), discountPercent, discountApproverId, pay (usdCash, khxCash, khqr, bank), change, fxRate, status, returns[]
Drawer shift	id, date, templateCode, register, cashierId, openedAt, closedAt, float, counts (first and recount), countedUSD, countedKHR, status, reviewedBy, note
Approval	id, type (void, return, discount, reopen), saleId, cashierId, reason, raisedAt, status, approverId, mode (on-site or queued), decidedAt
Cash movement	id, type (float, drop, payout, pay-in, deposit), shiftId, USD, KHR, by, receivedBy, status, reason
Stock movement	id, sku, type, qty ±, reason, by, at, ref

14 Quality and UI standards

- All UI text in Khmer with Arabic numerals; no English in the interface except brand names (KHQR, Bakong).
- No native select, alert, confirm or tooltips; shared custom dialogs replace them.
- One emerald accent colour. Red and amber appear only on exceptions; normal states stay neutral grey. Light theme by default; dark mode on every page.

- Responsive, with an off-canvas sidebar below 1024 px; 48 px targets on the cashier screen; empty, loading and error states on every page.
- Lists remember their tab, filters, search and page, and highlight the record you return from. Growing lists are paged (25 rows by default; receipts 20, audit 50).
- View and create pages are full pages with a back button, not modals; one : menu per table row.
- **Testing is manual:** in a browser at desktop and phone width, in both themes, plus headless Chrome checks. There is no automated test suite yet.

15 Status, issues and backlog

15.1 Build status

Area	Status	Notes
Login, page guards, view switching	Done	Email or phone + password; PIN for approvals
Cashier core (C1–C16, C22)	Done	Blind count, riel change, open shift, hold, voids and returns, override, safe drop, lock, in-terminal KHQR, tax invoice
Customer display (C17)	Done	Idle messages set by the manager
Keyboard shortcuts (C19)	Done	F2, F4, F8, Esc
Manager pages (spec 02)	Done	All 11 pages and their sub-pages
Owner pages (spec 03)	Done	Including payroll, banks, Stock In costs
Stock, phases 1–3	Done	On main
Online/offline badge (C18)	Not built	Waits for the offline phase
Line discount and price override (C20)	Not built	The cart discount covers the prototype
No-sale drawer open (C21)	Not built	Needs a cash drawer
Automatic KHQR confirmation	Not built	Needs the Bakong API

15.2 Issues found while preparing this report

#	Severity	Issue	Cause and fix
1	Minor	The owner's stock-value table shows an empty «ទំងន់ · 0 មុខ · \$0.00» row above the real categories (figure 18).	<code>totalStockValueAtCost()</code> loops over CATEGORIES, which starts with the «all» filter tab. Skip id: 'all'.
2	Minor	Payroll for a period that starts before the 14-day demo history counts the days with no shifts as absences (for example 77 absent days and \$730 of deductions for 7 Sep – 6 Oct).	Demo only. Start the history at the period start, or don't count absences before the first generated day.
3	Docs	README.md is out of date: 4-digit PINs, a person-then-PIN login, role-coloured sidebars, stock described as static, a 3-person staff list.	Update from CLAUDE.md, which is current.
4	Docs	docs/planning/stock-management.md says phase 3 has not started and the branch is unmerged.	Mark phases 2–3 done.
5	Info	The manager spec still lists shift statuses in indigo and sky, which the colour rules no longer allow.	Align the spec with the built (neutral) chips.

15.3 Open decisions

The cashier backlog (docs/planning/cashier-improvements.md §5) lists decisions D1–D7, and the payroll spec lists D-P1 (whether the night premium of 200% applies to all night hours) and D-P3 (when cash shortages are deducted). Settle these with the owner and an accountant before the backend data model is fixed.

16 Production readiness, risks and plan

16.1 Gaps

Gap	What production needs	Priority
Server and database	One shared record for every till, manager and owner; business rules (approvals, stock, cash, payroll) enforced on the server; nothing lost when a browser is cleared	P0
Authentication	Hashed passwords and PINs checked on the server; sessions; device registration per register; password reset by the owner	P0
Bakong KHQR	A merchant account per bank; dynamic KHQR; automatic confirmation (MD5 check through the Bakong API) instead of the manual button	P0
Offline mode	A PWA with a local queue so the till keeps selling when the internet drops, then syncs without duplicates	P1
Hardware	ESC/POS thermal printing, cash drawer kick, second-screen customer display via a print bridge or native wrapper; scanners already work	P1
Compliance	An accountant's review of receipts and tax invoices against GDT rules; payroll rules (D-P1); data retention and backups	P1
Automated tests	Unit tests for money maths (totals, change, expected cash, variance, profit, payroll) and end-to-end tests for the daily flow	P1
Multi-branch	Branch selector, transfers, consolidated owner reports	P2

16.2 Risks

Risk	Impact	Mitigation
Browser checks mistaken for real security	High	Never use the prototype with real money; move all checks to the server first
KHQR fraud (a fake payment screenshot)	High	Automatic Bakong confirmation; never rely on the customer's phone
Internet outage at the counter	Medium	Offline queue (PWA) before the pilot
Staff resistance to blind counts	Medium	Training; show the variance trend improving; keep the one-recount rule
Labour-law interpretation of night pay	Medium	Accountant review (D-P1) before the first real payroll
Scope creep from eBMS (purchasing, accounting)	Medium	Keep the POS boundary: stock movements and sales hand off to eBMS

16.3 Suggested plan

Phase	Work	Outcome
0 · Tidy (days)	Fix issues 1–5; refresh README; freeze the scope	A clean, documented baseline
1 · Validate (2 weeks)	Usability sessions with 3–5 cashiers and a manager on the demo; settle D1–D7, D-P1, D-P3	An approved front-end specification
2 · Backend (6–8 weeks)	API and database for staff, catalogue, shifts, sales, approvals, cash, stock, payroll; move the logic in data.js, manager-data.js and admin-data.js to the server; real authentication	Tills share one record
3 · Payments and hardware (3–4 weeks)	Bakong integration; thermal printer and cash drawer; offline queue	A till ready for a real counter
4 · Pilot (2–4 weeks)	One shop; daily review of variances and exceptions; weekly fixes	A go or no-go decision for rollout

Durations are planning estimates for a small team and should be confirmed once the backend scope is fixed.

A Appendix

A.1 Page inventory

Area	Pages
Public	index.html (login) · 404.html
Cashier (5)	terminal/pos-terminal · display/customer-display · receipts/receipts · shift/open-shift · shift/close-shift
Manager (17)	dashboard · approvals + view-request · shifts + view-shift · roster · cash + create-movement · stock + create-stock-in + create-adjustment · stock-count + create-count · stock-history · exceptions · reports/sales-report · settings
Owner (12)	dashboard · reports/profit-report · reports/stock-value · reports/shrinkage · reports/payroll · staff + view-staff · products · stock/stock-in + view-stock-in · settings · audit

A.2 Demo accounts

Two example accounts: cashier «ច្បង មករា» uses 012 345 678 or makara@digitechkh.com with the password makara2026; owner «ហេង ចាន់ថា» uses 012 999 000 with chantha2026. Every account is listed on the login page under «ព័ត៌មានសម្រាប់គំរូសាកល្បង» (a click fills the form), next to the button that resets the demo data. PINs are in section 2.

A.3 Glossary

English	Khmer on screen	Meaning
Shift	វេន	A time window of work; also one drawer session
Register	បញ្ជីគិតលុយ	A till: POS-01 to POS-03
Float	ប្រាក់បាតថត	Cash placed in the drawer at opening
Safe drop	ផ្ទេរចូលទូរដែក	Moving excess cash from the drawer to the safe
Void	លុបចោល	Cancelling a completed sale
Return	ប្រគល់ទំនិញវិញ	Taking goods back and refunding
Held sale	ការលក់ល្អរ	A parked cart to finish later
Override	បញ្ចុះលើសកំណត់	A discount above the cashier's limit, approved by a manager
Blind count	រាប់ដោយមិនមើលចំនួនរំពឹងទុក	Counting cash or stock without seeing the expected figure
Variance	ភាពខុសគ្នា	Counted minus expected
Countersign	ចុះហត្ថលេខាត្រួតពិនិត្យ	The manager's review of a closed shift
Exceptions	ករណីមិនប្រក្រតី	Unusual events, compared per cashier

English	Khmer on screen	Meaning
Shrinkage	ការខាតបង់ស្តុក	Stock lost to damage, expiry, theft or count shortages
Audit log	កំណត់ហេតុសវនកម្ម	Read-only history of every change
Exchange rate	អត្រាប្តូរប្រាក់	Riel per dollar, set daily

A.4 Related documents

- docs/spec/01–04: cashier, manager, owner, and work time and payroll specifications
- docs/planning/: cashier backlog, stock management, shift scheduling, customer display
- docs/research/pos–market–research.md: market and Cambodia research
- docs/presentation/DIGITECHKH–POS–Overview.pdf and ...–Workflow.pdf: Khmer slide decks for shop owners

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